STAT452 Final Project: Wine Quality and Experimental Design

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Abstract

This report completes both components of the Group 8 final project. Part 1 predicts red Vinho Verde quality from 11 physicochemical measurements with a locked, leakage-controlled comparison of OLS, Ridge, Lasso, and Elastic Net. Part 2 uses an observational two-factor analysis of student mathematics scores by test preparation and lunch category, with diagnostics, robust sensitivity checks, and explicitly non-causal interpretation.

1 Introduction and analysis protocol

This project addresses two complementary questions. Part 1 asks how accurately red-wine sensory quality can be predicted from routine physicochemical measurements. Part 2 asks whether mathematics scores differ across observed test-preparation and lunch groups, including whether the two associations interact. The analyses follow the course brief's emphasis on justified choices, reproducibility, and honest limitations.

For Part 1, RMSE is primary because large one-wine prediction errors are costly; MAE expresses typical error directly in quality-score points, and R^2 measures improvement over the holdout mean benchmark. Model selection uses fixed five-fold cross-validation, while the stratified 20% holdout is opened once (Hastie et al. 2009).

2 Part 1: Wine Quality (Red)

3 Exploratory Data Analysis

3.1 Dataset, variables, and prediction question

The UCI Wine Quality data describe red Portuguese Vinho Verde samples using 11 continuous physicochemical measurements and a sensory score supplied as the median of at least three expert ratings (Cortez et al. 2009a, 2009b). The response is `quality`; the predictors cover acidity, sugar, chlorides, sulfur dioxide, density, pH, sulphates, and alcohol. All predictors are numeric, and the detailed data dictionary appears in Appendix A.

We model quality numerically because regression preserves its ordering and avoids an arbitrary good/poor threshold. This assumes that a one-point change has roughly comparable meaning across the observed 3–8 range. Gaussian regression can produce fractional expected scores, so the interpretation is prediction of expected expert score rather than exact ordinal classification.

After the integrity checks in Section 2, the unique records were divided by a fixed quality-stratified 80/20 split. Every plot below uses only the 1088 training observations; the 271 holdout outcomes were not inspected.

3.2 Response and predictor distributions

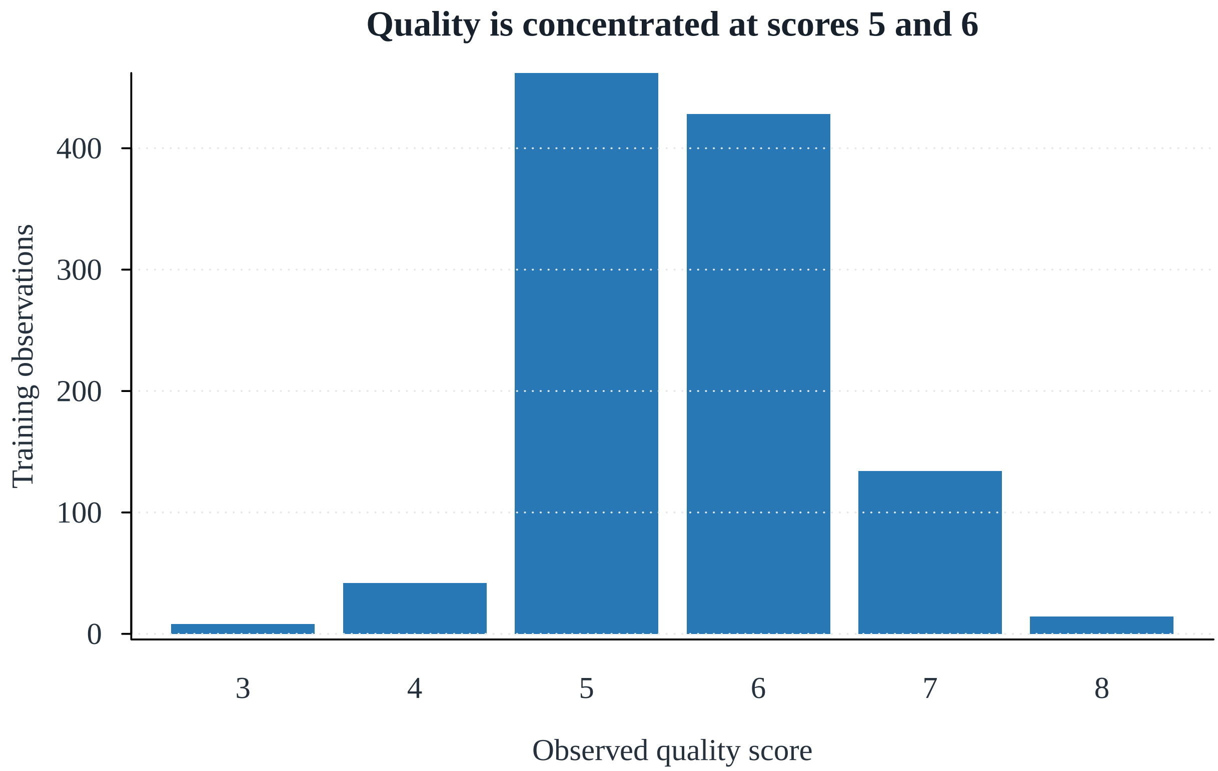

Figure 1: Training-set distribution of the sensory response. Scores 5 and 6 dominate, while the extremes contain few observations.

The response is concentrated at 5 and 6. A mean-like prediction is therefore a strong baseline, and error estimates for rare scores 3, 4, and 8 will be less precise. This imbalance also foreshadows regression toward the center.

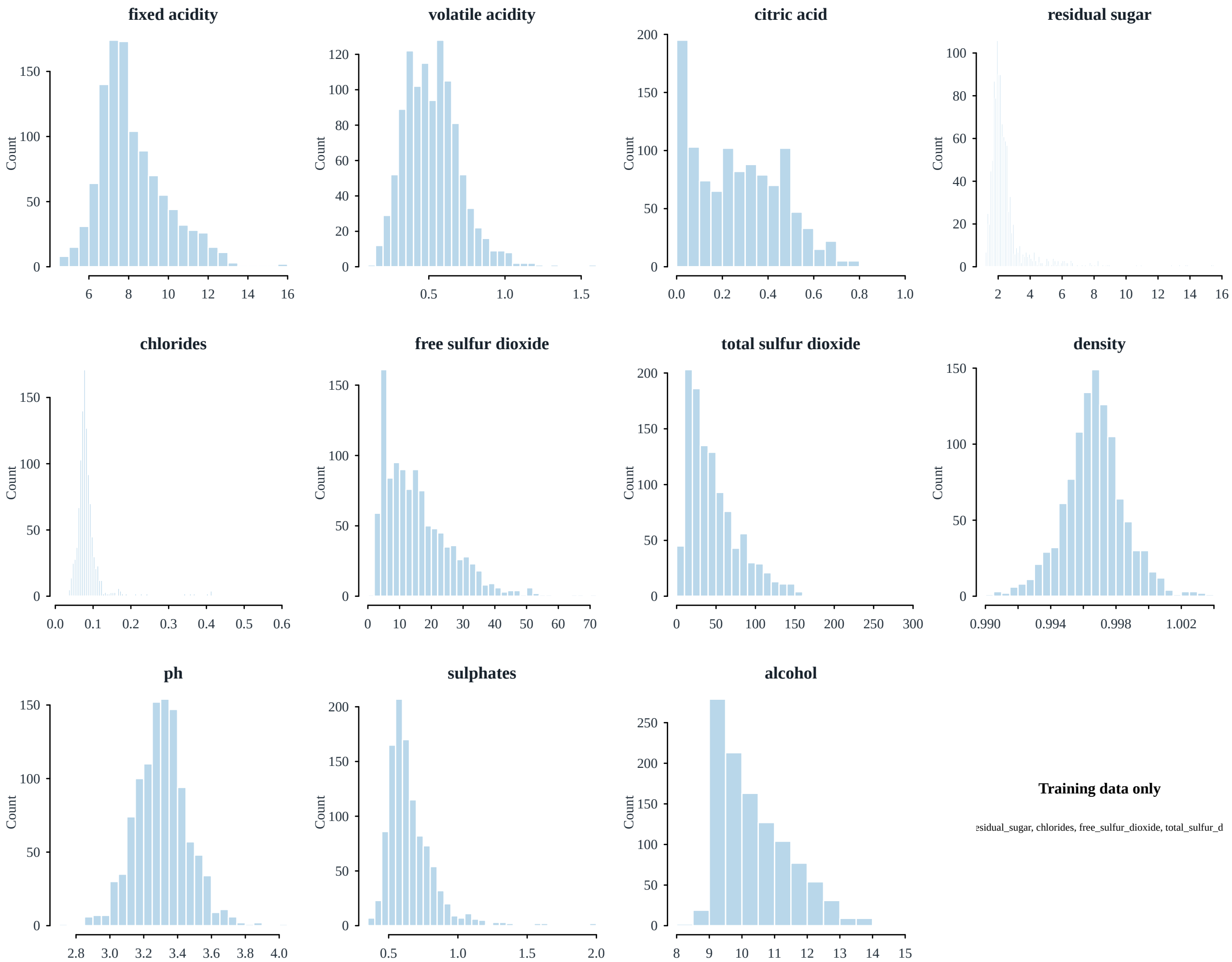

Figure 2: Training-set predictor distributions before transformation. Several chemistry measures are strongly right-skewed.

Residual sugar, chlorides, sulfur-dioxide measures, and sulphates have long right tails. These distributions motivate the transformations documented in Section 2; EDA itself is shown on the original measurement scale.

3.3 Pairwise relationships and correlation structure

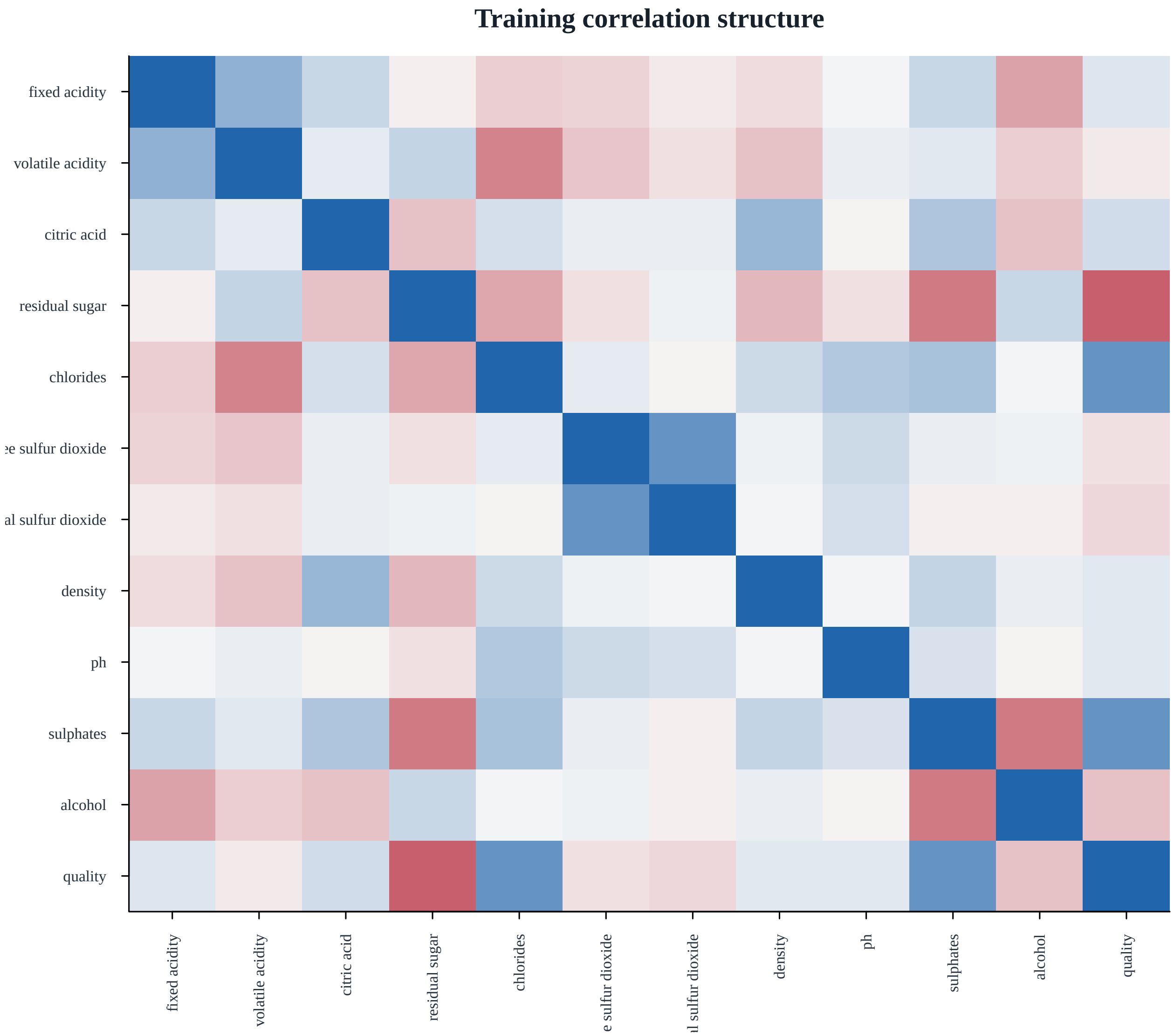

Figure 3: Training-set Pearson correlations. Blocks among acidity, density, pH, and sulfur-dioxide measures motivate coefficient stabilization.

The strongest marginal response signals are alcohol ($r = 0.47$); volatile acidity ($r = -0.37$); sulphates ($r = 0.23$); citric acid ($r = 0.22$). The heatmap also shows correlated predictor blocks, especially among acidity, density, pH, and the two sulfur-dioxide measurements. That structure makes OLS coefficients potentially unstable and directly motivates Ridge, Lasso, and Elastic Net.

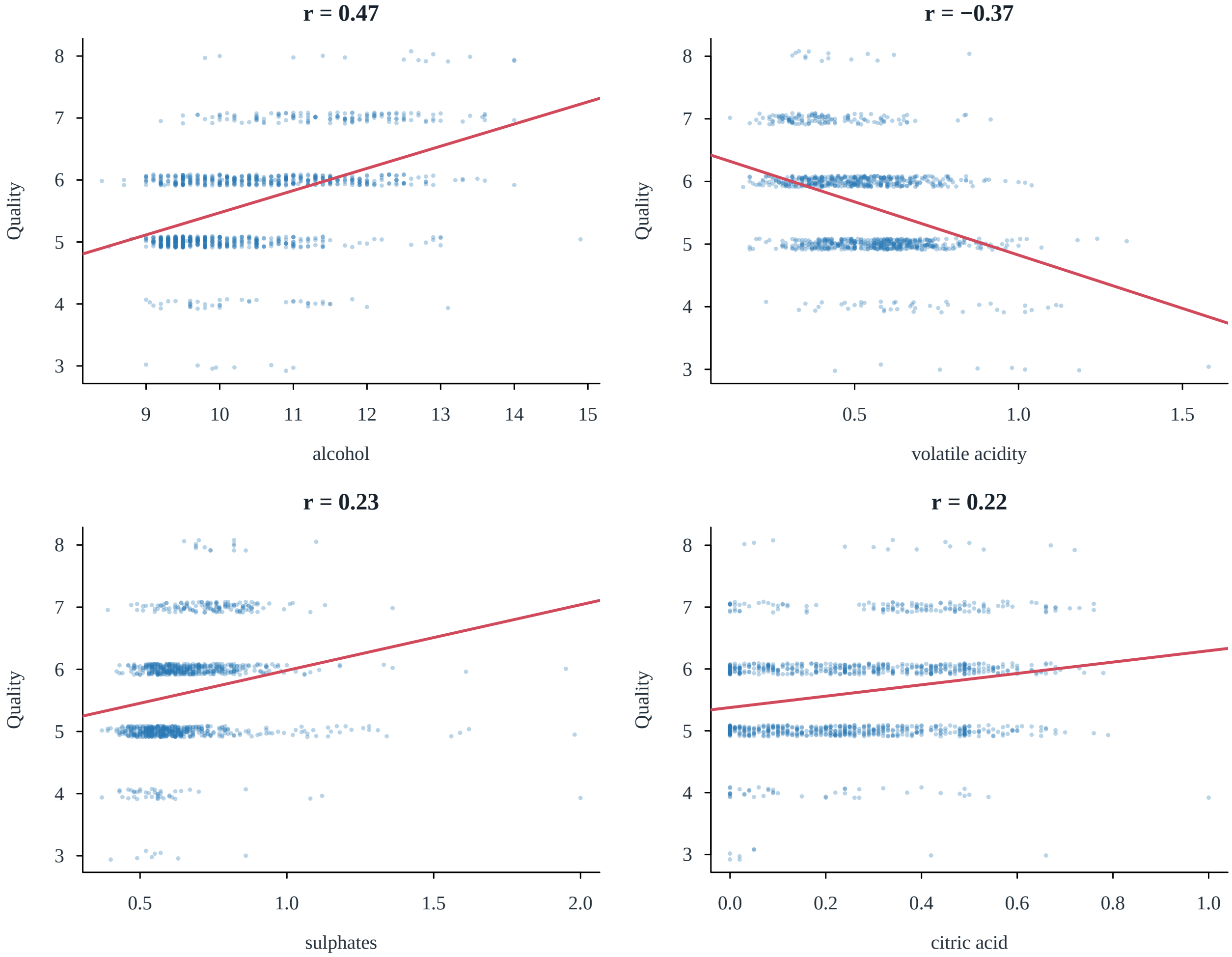

Figure 4: The four strongest marginal predictor–quality relationships in training. Jitter reveals the discrete response; lines are descriptive OLS trends.

Alcohol is positively associated with quality and volatile acidity negatively associated, but every score overlaps substantially on every predictor. No single physicochemical measurement is sufficient. These are predictive associations, not causal effects, because grape variety, brand, vintage, price, and production conditions are absent.

4 Data Cleaning

4.1 Missingness, duplicates, and the locked split

The raw archive contains 1599 rows, 11 numeric predictors, no missing cells, and no non-finite values. It also contains 240 exact duplicate records. Because the archive provides no sample identifier, an exact repeat could be either a copied row or a genuinely repeated batch. We conservatively keep the first copy and remove the rest **before** splitting, preventing identical profiles and labels from appearing in both training and holdout data. The limitation is that this choice may underweight genuinely repeated batches.

Table 1: Data-integrity audit and decisions.

Check	Result	Decision
Raw observations	1599	Validated against UCI metadata
Numeric predictors	11	All retained
Missing cells	0	None to impute
Non-finite numeric cells	0	None
Exact duplicate rows removed	240	Remove before splitting to prevent duplicate leakage
Unique observations	1359	Analysis population
Observed quality range	3–8	Treat as numeric regression response

The fixed seed 4520803 produces a quality-stratified 80/20 partition after deduplication. Seed 4520804 then assigns five stratified folds inside training. The shared fold IDs are reused by every model (Hastie et al. 2009).

Table 2: Locked data partitions after deduplication.

Partition	Observations	Percent_of_unique
Raw archive	1599	NA
Unique before split	1359	100.0
Training	1088	80.1
Holdout	271	19.9

4.2 Outliers and transformations

Tukey's 1.5-IQR rule (Tukey 1977) flags 277 training rows (25.5%). Deleting all of them would discard a large and scientifically plausible part of the target population; the dataset metadata also warns that rare wines are expected. We therefore retain every row, winsorize each predictor at its training 1st and 99th percentiles to limit leverage, and never clip the response.

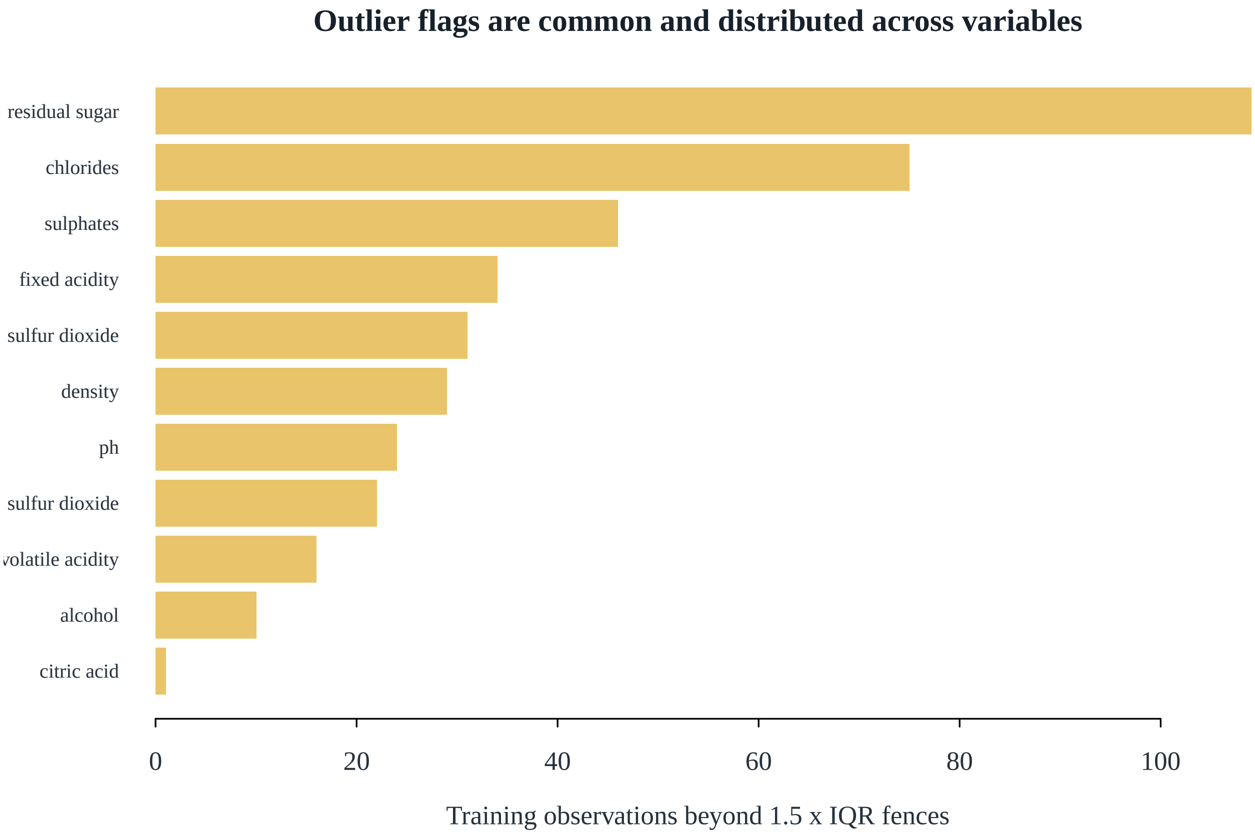

Figure 5: Training observations flagged by the univariate Tukey-IQR rule. Flags are diagnosed and retained rather than deleted wholesale.

The training skewness rule selects residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, sulphates for `log1p` transformation. Median imputation is defined as a reproducibility safeguard but is a no-op for this dataset. After winsorization and optional `log1p`, all predictors are centered and scaled so penalty magnitudes are comparable.

4.3 Leakage control

All medians, winsorization limits, transformation choices, means, and standard deviations are learned without holdout outcomes. During cross-validation, those quantities are refit inside each analysis fold and applied to its validation fold. The final recipe is fit once to all training predictors and then applied unchanged to the predictor-only holdout file. This design prevents preprocessing information from leaking backward into feature selection or tuning.

5 Feature Selection

5.1 Screening without deleting correlated information

We use three training-only views of relevance: marginal correlations, variance inflation factors (VIFs) (Fox and Weisberg 2019), and an AIC sensitivity analysis (Akaike 1974). The largest VIF is 7.90 for fixed acidity. This is material multicollinearity but not evidence that the feature is useless: deleting one correlated chemistry measure can make the retained coefficient depend arbitrarily on that choice. Therefore all predictors enter Ridge, while sparse selection is delegated to Lasso's embedded penalty.

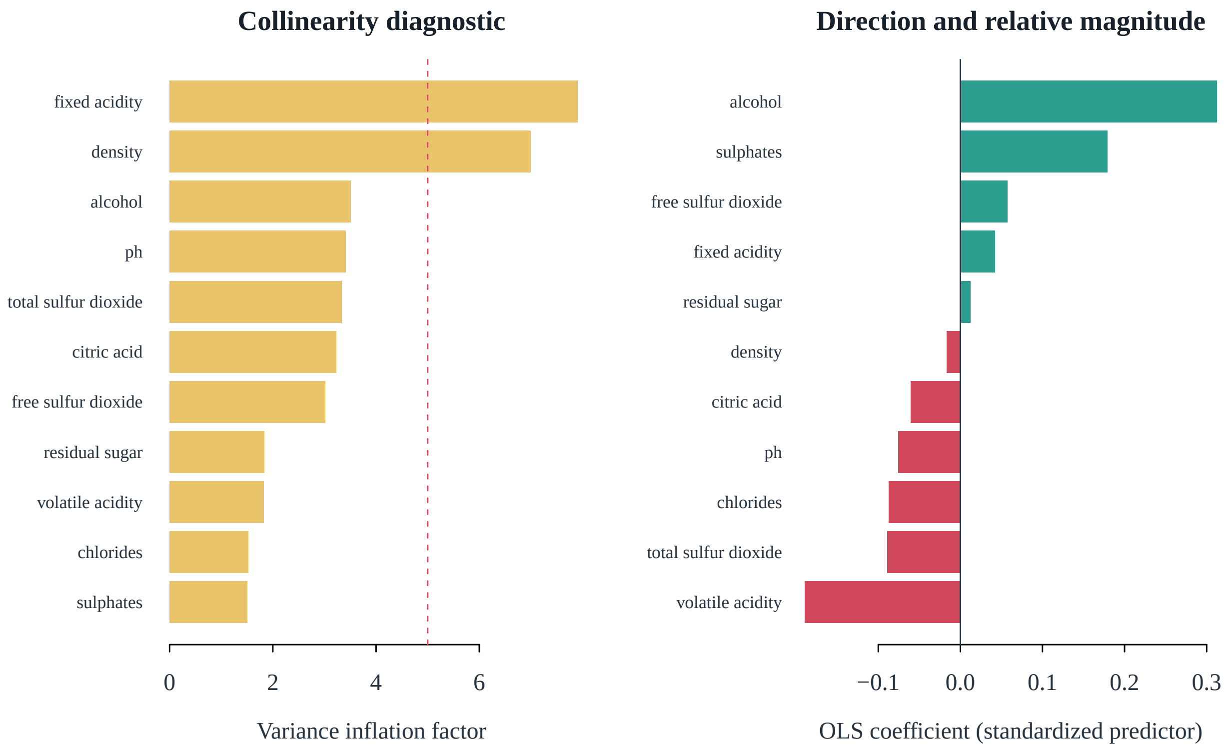

Figure 6: VIFs and full-OLS coefficients after the training-derived transformations. Coefficients are comparable because predictors are standardized.

The AIC sensitivity model retains 7 of 11 predictors: alcohol, volatile acidity, sulphates, chlorides, ph, total sulfur dioxide, free sulfur dioxide. We do not promote this data-dependent subset directly to the holdout comparison; it is a diagnostic against which to compare Lasso.

5.2 Mean and OLS baselines

Table 3: Mean and OLS baseline performance before holdout evaluation.

Model	Training_RMSE	CV_RMSE	CV_MAE	CV_R2	Predictors
Mean	0.824	0.824	0.695	0.000	0
OLS	0.663	0.672	0.516	0.335	11

The mean-only model establishes the no-chemistry benchmark. OLS lowers cross-validated error, but correlated predictors allow coefficient variance. Regularization deliberately accepts some training bias to reduce that variance.

For influence sensitivity, the training OLS fit flags 73 rows above the conventional Cook's-distance $4/n$ reference rule; the maximum is 0.0314. These flags identify cases worth scrutiny rather than automatic deletion. A holdout row cannot influence the already fitted coefficients, so held-out diagnostics focus instead on unusually large prediction errors.

5.3 Embedded Lasso selection

At the one-standard-error penalty, Lasso retains 7 predictors: alcohol, volatile acidity, sulphates, chlorides, ph, total sulfur dioxide, fixed acidity. This is the required principled feature-selection result. The one-SE rule is used for interpretation because it favors a stable sparse description; the minimum-CV penalty remains the predeclared predictive fit.

Table 4: Embedded Lasso feature selection using the one-standard-error penalty.

Feature	Coefficient	Selected
alcohol	0.2996	Yes
volatile_acidity	-0.1664	Yes
sulphates	0.1253	Yes
chlorides	-0.0360	Yes
ph	-0.0177	Yes
total_sulfur_dioxide	-0.0136	Yes
fixed_acidity	0.0038	Yes
citric_acid	0.0000	No
residual_sugar	0.0000	No
free_sulfur_dioxide	0.0000	No
density	0.0000	No

6 Modelling with Regularization

For standardized transformed predictors, all regularized candidates minimize

$$\frac{1}{2n}\|\mathbf{y} - \beta_0 - \mathbf{X}\beta\|_2^2 + \lambda\left\{\alpha\|\beta\|_1 + \frac{1-\alpha}{2}\|\beta\|_2^2\right\}.$$

Ridge uses $\alpha = 0$ (Hoerl and Kennard 1970), Lasso $\alpha = 1$ (Tibshirani 1996), and Elastic Net searches $\alpha \in \{0.25, 0.50, 0.75\}$ (Zou and Hastie 2005). For each alpha, 120 decreasing λ values from 100 to 10^{-4} use identical folds (Hastie et al. 2009). Coordinate-descent fits use `glmnet` (Friedman et al. 2010).

6.1 Cross-validation and shrinkage

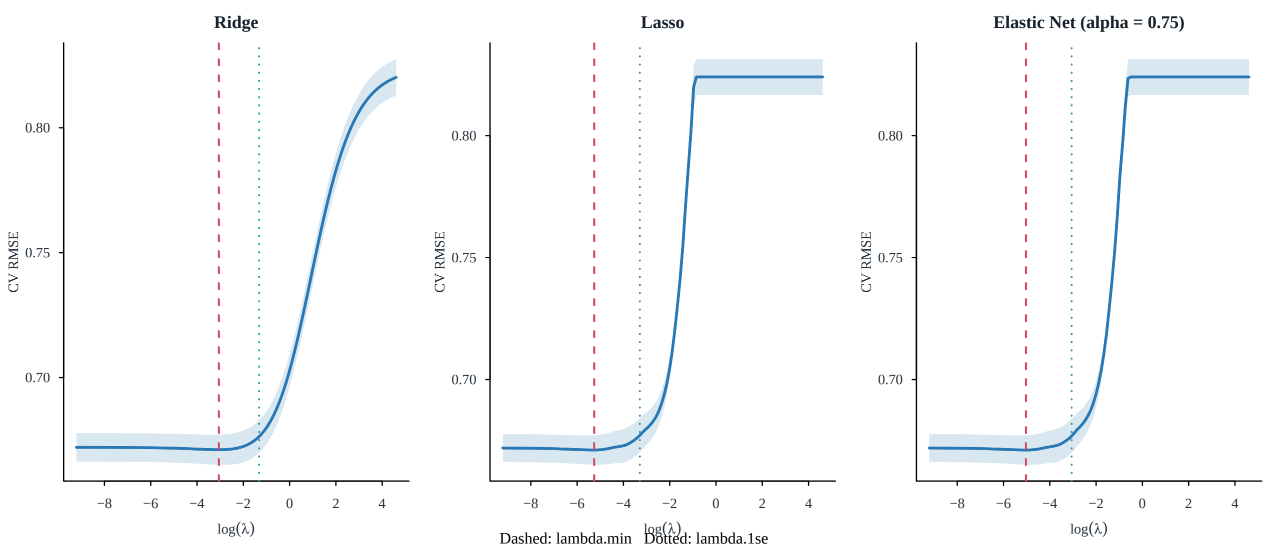

Figure 7: Five-fold training CV curves with fold-specific preprocessing. Dashed lines minimize CV error; dotted lines apply the one-standard-error rule.

Elastic Net selected $\alpha = 0.75$. The minimum-error and one-SE penalties, CV errors, and effective model sizes are reported below. Ridge keeps all coefficients but continuously shrinks them; Lasso and Elastic Net can set coefficients exactly to zero.

Table 5: Regularized-model tuning and effective size on the shared training folds.

Model	Alpha	Lambda_min	Lambda_1se	CV_RMSE_min	Nonzero_min	Nonzero_1se
Ridge	0.00	0.0470	0.2683	0.6711	11	11
Lasso	1.00	0.0052	0.0373	0.6712	9	7
Elastic Net	0.75	0.0065	0.0470	0.6712	9	7

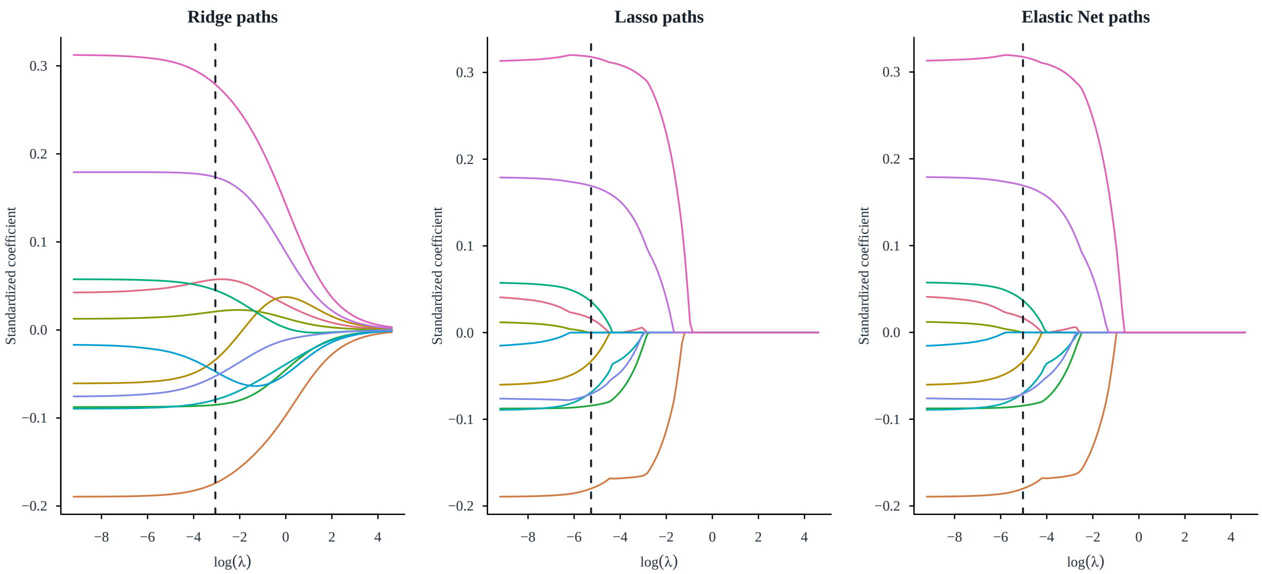

Figure 8: Coefficient paths as regularization changes. The vertical line in each panel is the minimum-CV penalty used for prediction.

As λ grows, paths move toward zero. Lasso reaches exact zeros earliest; Ridge shares weight among correlated measurements; Elastic Net interpolates. This is the expected bias–variance mechanism rather than merely a software choice.

6.2 Pre-holdout decision

Before reconstructing any holdout outcome, we locked **Ridge**, whose five-fold CV RMSE was 0.6711. This written lock prevents choosing a preferred model after seeing test performance. All predeclared models are nevertheless evaluated to make the comparison transparent.

7 Evaluation

7.1 Fair model comparison

Table 6: Fair holdout comparison; the model lock was based only on five-fold CV RMSE.

Model	CV_RMSE	Holdout_RMSE	RMSE_CI_Lower	RMSE_CI_Upper	Holdout_MAE	Holdout_R2	Locked_by_CV
OLS	0.6721	0.6131	0.5565	0.6677	0.4803	0.4406	No
Ridge	0.6711	0.6134	0.5571	0.6712	0.4821	0.4400	Yes
Elastic Net	0.6712	0.6139	0.5582	0.6703	0.4822	0.4391	No
Lasso	0.6712	0.6139	0.5582	0.6703	0.4822	0.4390	No
Mean	0.8242	0.8198	0.7461	0.9018	0.6923	-0.0001	No

On the same 271 rows, the pre-locked Ridge obtained RMSE 0.613, MAE 0.482, and R^2 0.440. The observed lowest holdout RMSE belongs to **OLS** (0.613). We retain the CV lock as the confirmatory choice even if this ranking differs: selecting by the holdout would make its error optimistic. Fixed-seed nonparametric bootstrap RMSE intervals (Efron and Tibshirani 1993) show that small ranking differences are not decisive.

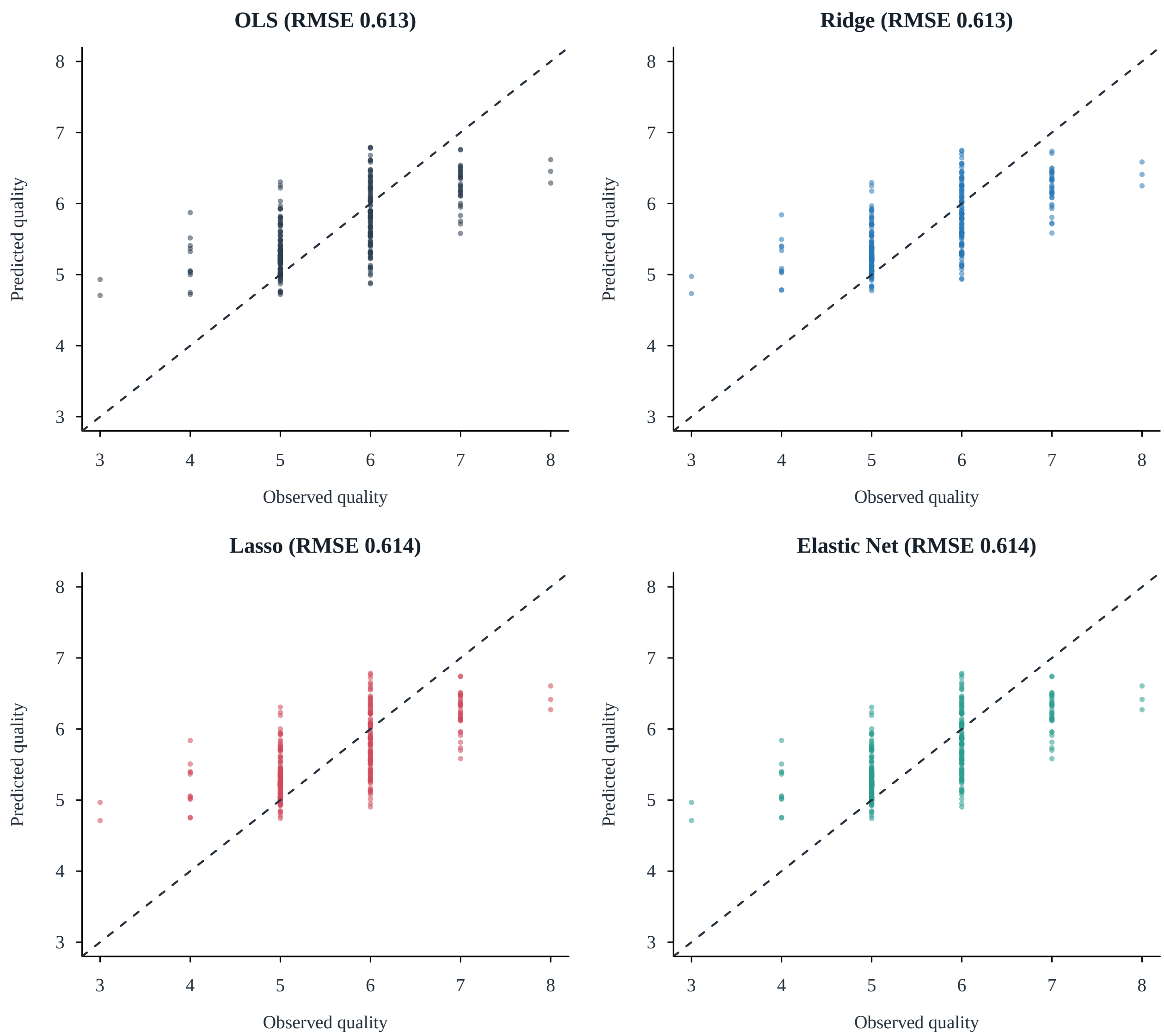

Figure 9: Held-out predictions from the four fitted regression models. The diagonal denotes perfect prediction.

Predictions shrink toward the common scores 5–6, which lowers average error but systematically misses rare extremes. This pattern follows from response imbalance and squared-error fitting, not simply from regularization.

7.2 Diagnostics and bias–variance trade-off

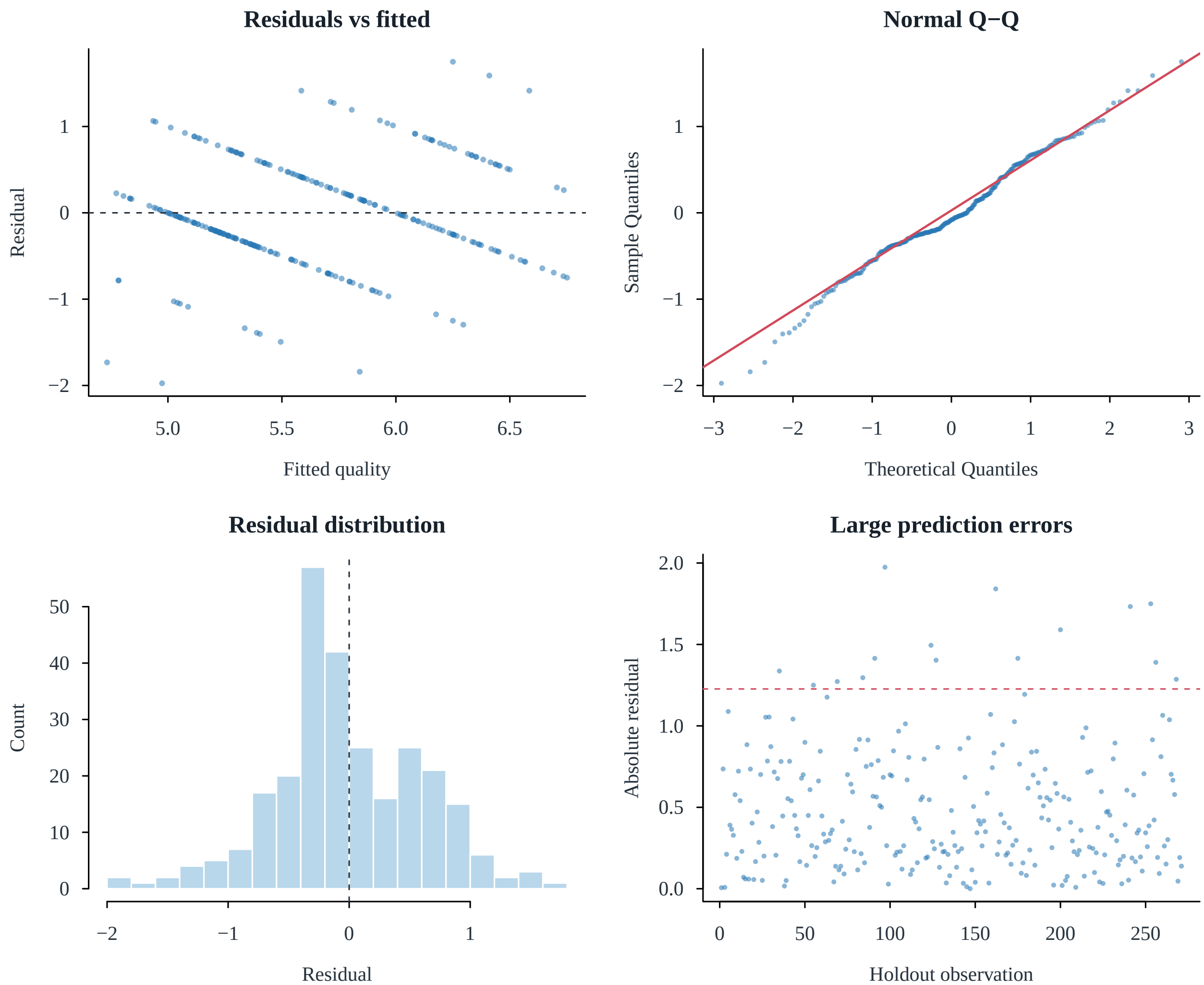

Figure 10: Residual-shape and absolute-error checks for the pre-locked model on the untouched holdout set. The dashed line in the final panel is twice the model RMSE. These diagnostics describe generalization and were not used to retune the model.

OLS has the least shrinkage and therefore the lowest regularization bias but the greatest sensitivity to correlated predictors. Ridge reduces coefficient variance without selection; Lasso trades additional bias for sparsity; Elastic Net stabilizes groups of correlated features while retaining some selection. The relevant empirical comparison is the gap among training, CV, and holdout RMSE in Table 6. The regularized candidates have similar held-out accuracy, so stability and interpretability matter more than a small decimal-place difference.

7.3 Limitations

1. Quality is ordinal and discrete; Gaussian regression treats adjacent scores as equally spaced and permits fractional predictions.
2. Rare scores 3, 4, and 8 yield imprecise tail performance, while the bootstrap quantifies only sampling variation in this holdout.
3. Exact deduplication prevents leakage but may remove genuinely repeated batches because the archive has no sample identifier.
4. Physicochemical measurements omit brand, vintage, grape variety, price, and production process; predictive associations are not causal effects.

5. Winsorization limits leverage and log transforms reduce skew, but alternative robust losses or ordinal models could change tail behavior.

7.4 Conclusion

The full workflow meets the Part 1 objective: documented EDA and cleaning, principled selection, a baseline plus three regularized models, shared-fold CV, coefficient-path interpretation, and a single locked holdout evaluation. Ridge is the confirmatory recommendation for expected-score prediction under this design. Alcohol and volatile acidity provide the strongest marginal signals, but correlated chemistry measurements and irreducible sensory variation limit accuracy. A future extension should compare ordinal regression or robust nonlinear models under the same locked resampling protocol.

8 Part 2: Experimental-design extension

8.1 Dataset, design, and hypotheses

The Timothy Adeyemi Kaggle listing for *Students Performance in Exams* contains 1,000 student records and eight variables (Adeyemi, n.d.). Its data card does not document random assignment, school/classroom clustering, or a sampling frame. We therefore describe this as an **observational (2×2) factorial analysis**, not a randomized experiment. All 1,000 rows are retained; there are no missing values or exact duplicate profiles, so neither imputation nor subsampling is needed.

The pre-specified response is mathematics score (0–100). Factor A is test preparation (`none`, `completed`), and Factor B is lunch category (`free/reduced`, `standard`). These variables meet the brief's requirement of two categorical factors and a continuous response. Lunch is interpreted as a contextual socioeconomic proxy, not a manipulable treatment or a label of individual ability. Reading and writing scores are parallel outcomes and are not used as covariates because conditioning on simultaneous outcomes would change the research question. Gender, race/ethnicity group, and parental education are not additional factors in the primary model: adding them would replace the required focused two-factor question with a sparse, sensitive multi-factor analysis. Their omission also means the two reported associations remain vulnerable to confounding.

The study asks whether mean mathematics score is associated with preparation, with lunch group, and with their interaction. At $\alpha = 0.05$, the three null hypotheses are: (i) equal preparation marginal means, (ii) equal lunch marginal means, and (iii) equal preparation differences across lunch groups. The alternatives are two-sided. Because assignment was not randomized, rejecting a null supports an adjusted group association within this factorial model, not a causal effect.

8.2 Descriptive statistics and visual evidence

Table 7: Math-score summaries for the four factorial cells.

Cell	n	Mean	Median	SD	Min	Max
free/reduced | none	224	56.51	58	15.09	0	93
standard | none	418	68.13	68	13.63	19	100
free/reduced | completed	131	63.05	64	14.43	23	100
standard | completed	227	73.53	73	13.02	32	100

The four cells contain between 131 and 418 students. This imbalance motivates effect-coded marginal tests rather than order-dependent sequential sums of squares. Within-cell standard deviations range from 13.02 to 15.09 points. Completed preparation and standard lunch are each descriptively associated with higher means, while substantial score overlap remains in every cell.

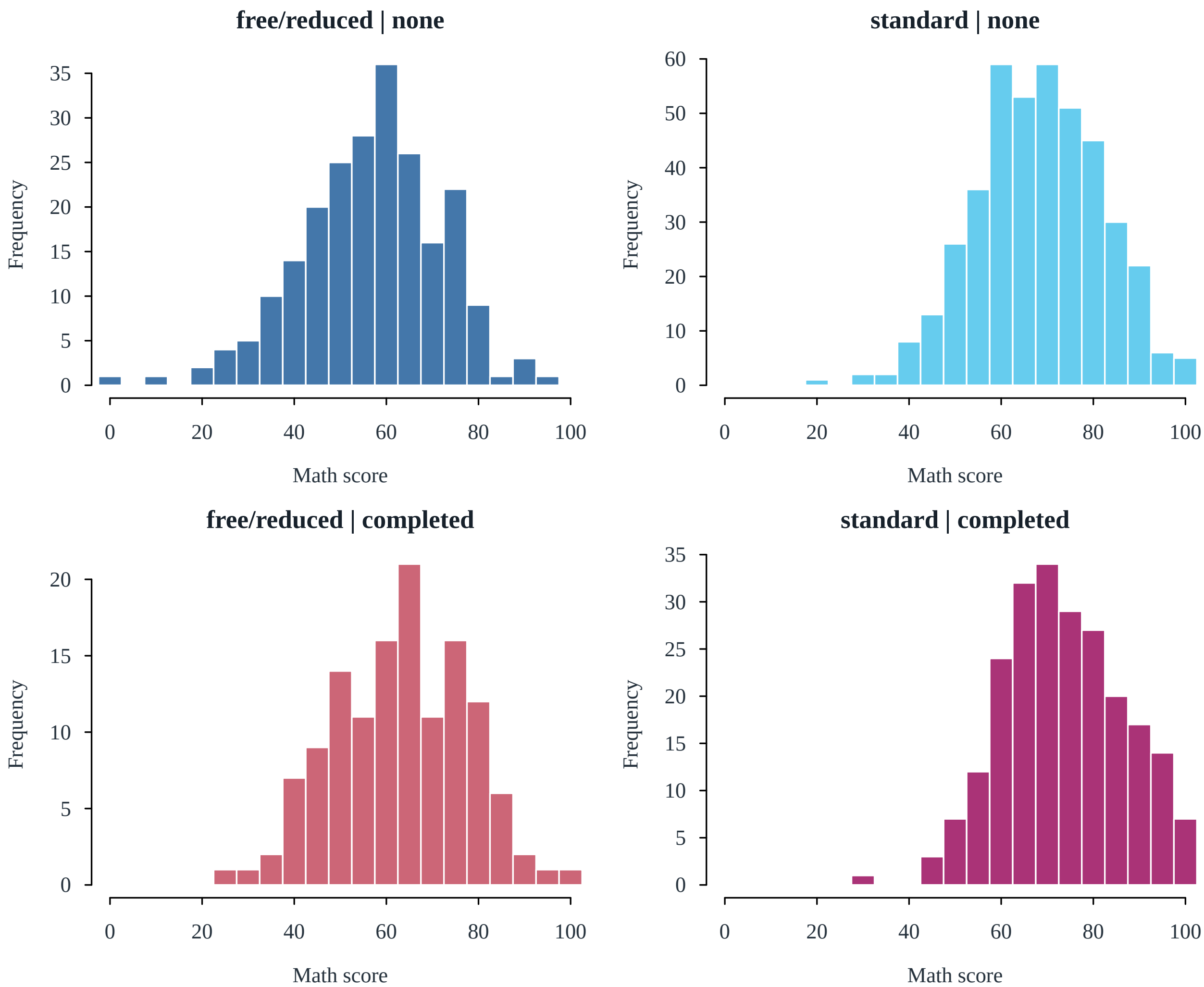

Figure 11: Within-cell mathematics-score histograms. Scores are bounded and discrete, with modest departures from Gaussian shape but no empty or extremely small cell.

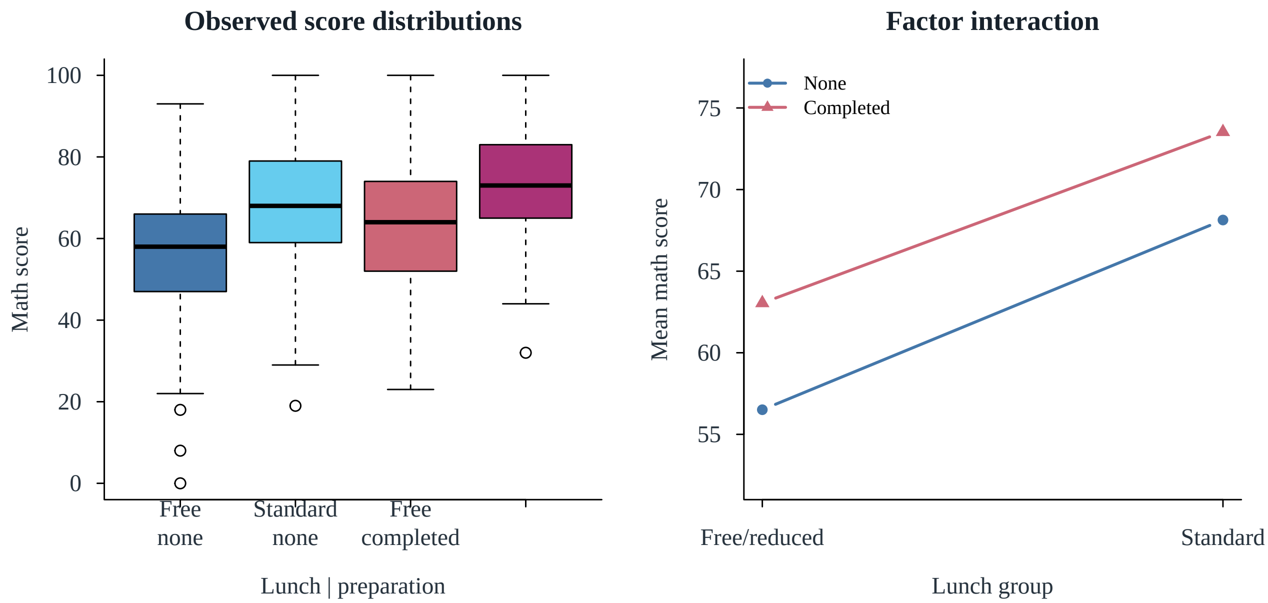

Figure 12: Boxplots and the mean interaction profile for lunch and test preparation. Near-parallel mean lines suggest a weak interaction relative to the two marginal differences.

8.3 Factorial ANOVA and effect sizes

We fit

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk},$$

where Y is mathematics score. Sum-to-zero contrasts make the main effects comparisons of equally weighted marginal cell means despite unequal sample sizes. The one-degree-of-freedom partial tests are Type-III-style tests from the full interaction model. Alongside (F) tests, we report score-point differences, 95% confidence intervals, and partial η^2 (Lakens 2013).

Table 8: Two-factor ANOVA results with point-scale and standardized effect sizes.

Effect	Estimate (95% CI)	F	p	Partial eta2
Test-preparation association	5.97 [4.09, 7.85]	38.791	<0.001	0.037
Lunch-group association	11.06 [9.18, 12.94]	133.132	<0.001	0.118
Preparation-by-lunch interaction	-1.14 [-4.90, 2.62]	0.352	0.553	0.000

Students who completed preparation averaged 5.97 points higher after equally weighting the lunch cells (95% CI [4.09, 7.85]). The standard lunch group averaged 11.06 points higher than the free/reduced group (95% CI [9.18, 12.94]). Both marginal nulls are rejected. The difference in preparation effects is -1.14 points and is not significant ($p = 0.553$); the data do not support effect modification by lunch category.

8.4 Diagnostics, model adequacy, and follow-up comparisons

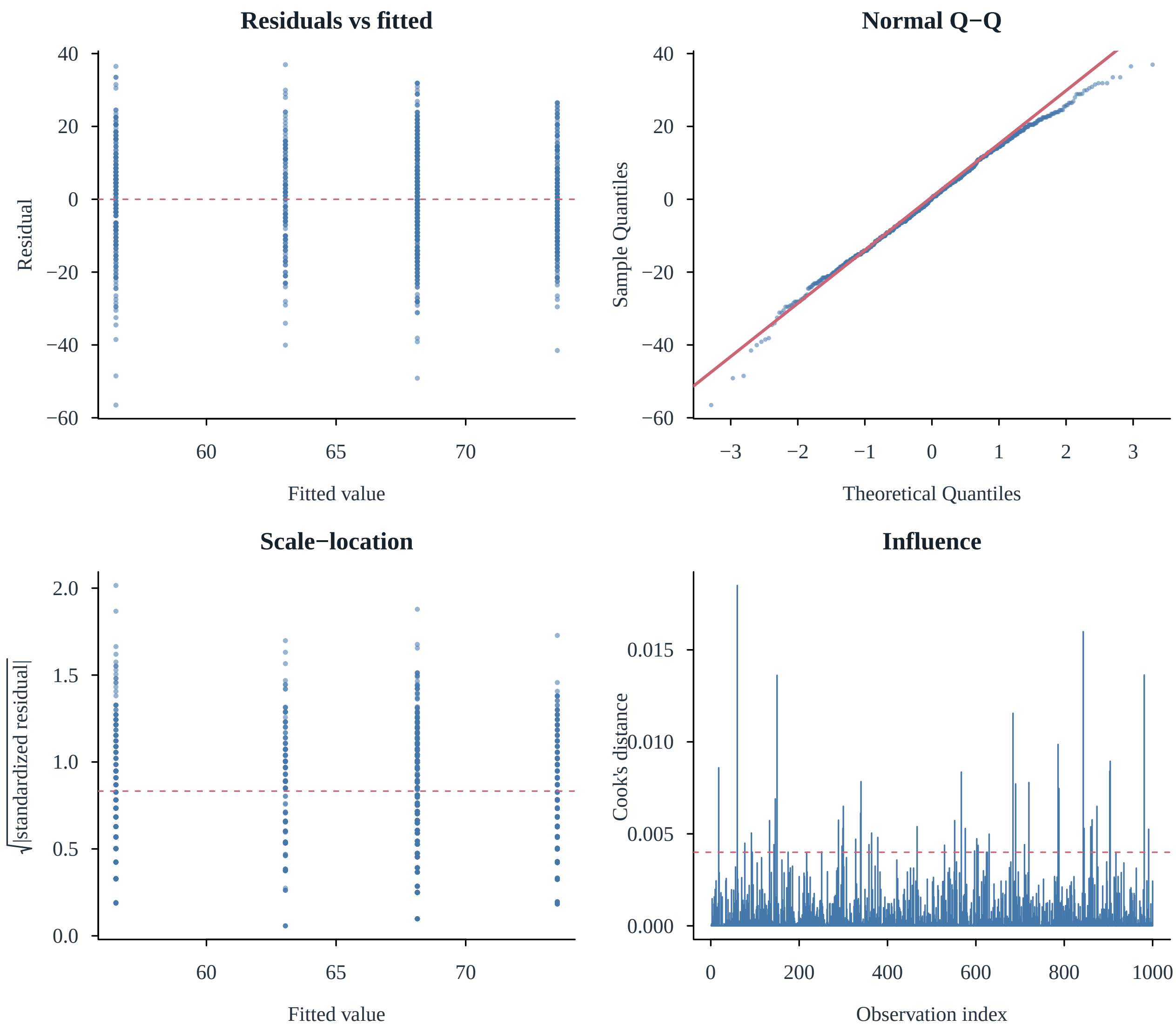

Figure 13: Residual, normal Q–Q, scale-location, and Cook's-distance diagnostics for the two-factor model.

The Shapiro–Wilk test (Shapiro and Wilk 1965) detects a modest non-Gaussian departure ($W = 0.996$, $p = 0.005$); at $n = 1000$, this test is sensitive to the discreteness and bounds of exam scores. The Brown–Forsythe test (Brown and Forsythe 1974) does not reject equal within-cell variances ($F = 1.06$, $p = 0.364$). The conventional $4/n$ influence rule flags 48 observations, but the maximum Cook's distance is only 0.0185. HC3 robust standard errors preserve the same decisions for both marginal effects and the interaction (MacKinnon and White 1985). Independence remains unverifiable because school and classroom identifiers are absent; this is the main model-adequacy limitation.

Each factor has only two levels, so a within-factor Tukey procedure would merely repeat its single main comparison. We instead report four scientifically direct simple contrasts and control their familywise error with Holm's method (Holm 1979).

Table 9: Follow-up simple effects with Holm-adjusted p-values.

Contrast	Estimate	Holm p
Preparation effect | free/reduced lunch	6.537	<0.001
Preparation effect | standard lunch	5.399	<0.001
Lunch effect | no preparation	11.625	<0.001
Lunch effect | completed preparation	10.487	<0.001

All four simple contrasts remain significant after adjustment, but they are secondary descriptions; the non-significant interaction remains the formal test of whether the preparation association differs by lunch group.

8.5 Practical conclusion and design evaluation

The lunch contrast is the larger observed association: 11.06 score points with partial $\eta^2 = 0.118$, compared with 5.97 points and partial $\eta^2 = 0.037$ for preparation. The interaction is practically small (partial $\eta^2 = 0.0004$) and its confidence interval includes differences in either direction. Reporting raw points as well as standardized effects keeps statistical significance separate from educational importance.

These findings should guide a future experiment, not be presented as evidence that changing lunch category or enrolling a particular student in preparation will cause the reported score change. Preparation participation may reflect motivation or prior attainment, and lunch category may proxy household and school resources. Income, prior scores, attendance, school quality, and classroom clustering are unavailable. The Kaggle listing also provides no collection protocol sufficient to establish representativeness.

An actionable next study would randomize access to a defined preparation program, block randomization by baseline score and lunch category, record school/class identifiers, and analyze score change from a pre-test. For the current data, the defensible conclusion is narrower: preparation completion and standard lunch are associated with higher mean mathematics scores, with no evidence that the preparation difference varies between lunch groups. All calculations and figures were generated in R (R Core Team 2026) through the canonical Stage 5 pipeline.

9 Contributions and review

The repository history, module ownership, and completed handoffs record the following contributions:

- **Nguyễn Đình Thiên Lộc:** project framework and reproducibility design; Part 1 data preparation/EDA; Part 2 data audit, design justification, and hypotheses.
- **Trần Lê Anh Tuấn:** Part 1 cleaning and preprocessing; Part 2 descriptive statistics and visualizations.
- **Lê Minh Thuận:** Part 1 feature selection and baselines; Part 2 effect-coded factorial ANOVA and effect sizes.
- **Nguyễn Bảo Minh Triết:** Part 1 regularized modelling and canonical integration; Part 2 interpretation and report integration.
- **Nguyễn Hồng Tấn Tài:** Part 1 locked holdout evaluation; Part 2 diagnostics, robust sensitivity, and multiplicity-adjusted follow-up comparisons.

The final integration review checked every code-to-table/figure handoff, the holdout boundary, contrast directions, diagnostics, citations, and PDF layout. Dataset discovery is not counted as a contribution.

A Detailed data documentation

Table 10: Wine Quality (Red) data dictionary.

Variable	Role	Type	Description
fixed_acidity	Predictor	Numeric	Fixed acidity (g/dm3)
volatile_acidity	Predictor	Numeric	Volatile acidity (g/dm3)
citric_acid	Predictor	Numeric	Citric acid (g/dm3)
residual_sugar	Predictor	Numeric	Residual sugar (g/dm3)
chlorides	Predictor	Numeric	Chlorides (g/dm3)
free_sulfur_dioxide	Predictor	Numeric	Free sulfur dioxide (mg/dm3)
total_sulfur_dioxide	Predictor	Numeric	Total sulfur dioxide (mg/dm3)
density	Predictor	Numeric	Density (g/cm3)
ph	Predictor	Numeric	pH
sulphates	Predictor	Numeric	Potassium sulphate (g/dm3)
alcohol	Predictor	Numeric	Alcohol (% by volume)
quality	Response	Integer score	Median sensory score from wine experts (0–10 documented; 3–8 observed)

Table 11: Training-set descriptive statistics.

Variable	N	Mean	SD	Min	Q1	Median	Q3	Max	Skewness
fixed_acidity	1088	8.259	1.713	4.600	7.100	7.850	9.100	15.900	0.955
volatile_acidity	1088	0.529	0.181	0.120	0.390	0.520	0.640	1.580	0.755
citric_acid	1088	0.271	0.194	0.000	0.090	0.260	0.420	1.000	0.321
residual_sugar	1088	2.558	1.445	1.200	1.900	2.200	2.600	15.500	4.453
chlorides	1088	0.089	0.052	0.012	0.070	0.079	0.091	0.611	5.448
free_sulfur_dioxide	1088	16.156	10.638	1.000	8.000	14.000	22.000	72.000	1.265
total_sulfur_dioxide	1088	47.348	33.716	6.000	22.000	38.000	64.250	289.000	1.569
density	1088	0.997	0.002	0.990	0.996	0.997	0.998	1.004	0.052
ph	1088	3.312	0.155	2.740	3.210	3.310	3.400	4.010	0.196
sulphates	1088	0.661	0.176	0.370	0.550	0.620	0.730	2.000	2.534
alcohol	1088	10.428	1.088	8.400	9.500	10.100	11.100	14.900	0.876
quality	1088	5.625	0.825	3.000	5.000	6.000	6.000	8.000	0.198

B Cleaning and outlier audit

Table 12: Training-set Tukey-IQR outlier audit; rows are retained.

Variable	IQR_Flags	Percent	Lower_Fence	Upper_Fence
fixed_acidity	34	3.125	4.100	12.100
volatile_acidity	16	1.471	0.015	1.015
citric_acid	1	0.092	-0.405	0.915
residual_sugar	109	10.018	0.850	3.650
chlorides	75	6.893	0.039	0.122
free_sulfur_dioxide	22	2.022	-13.000	43.000
total_sulfur_dioxide	31	2.849	-41.375	127.625
density	29	2.665	0.992	1.001
ph	24	2.206	2.925	3.685
sulphates	46	4.228	0.280	1.000
alcohol	10	0.919	7.100	13.500

Table 13: Training-only transformation and winsorization plan.

Variable	Raw_Skewness	IQR_Flags	Winsor_Lower	Winsor_Upper	Transformation
fixed_acidity	0.9547	34	5.1000	12.9000	standardize
volatile_acidity	0.7550	16	0.2000	1.0269	standardize
citric_acid	0.3214	1	0.0000	0.7013	standardize
residual_sugar	4.4527	109	1.4000	8.6260	log1p + standardize
chlorides	5.4480	75	0.0420	0.3888	log1p + standardize
free_sulfur_dioxide	1.2646	22	3.0000	51.0000	log1p + standardize
total_sulfur_dioxide	1.5691	31	8.0000	147.0000	log1p + standardize
density	0.0522	29	0.9919	1.0014	standardize
ph	0.1959	24	2.9387	3.7100	standardize
sulphates	2.5345	46	0.4300	1.3126	log1p + standardize
alcohol	0.8758	10	9.0000	13.4130	standardize

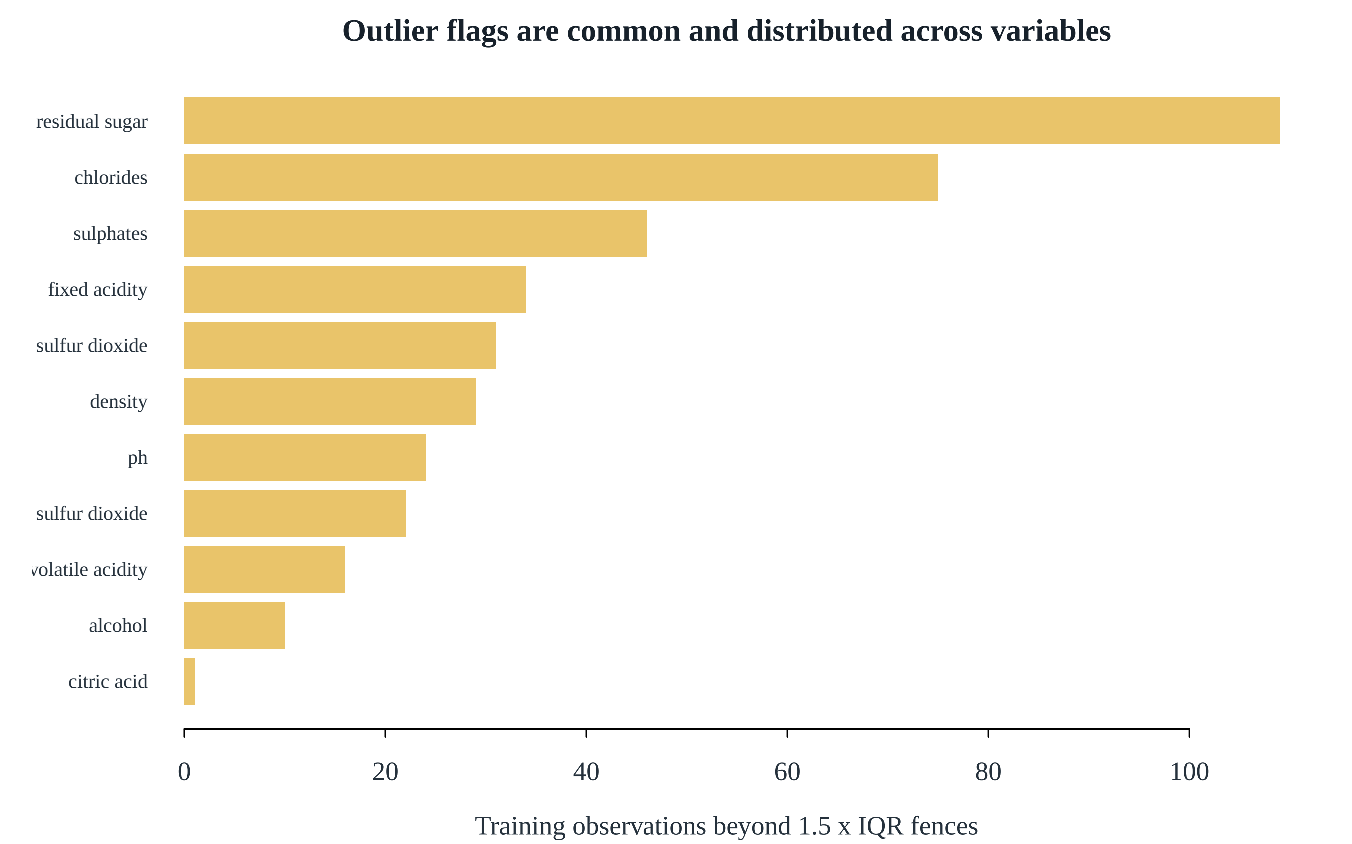

Figure 14: Training observations flagged by the univariate Tukey-IQR rule. Flags are diagnosed rather than deleted wholesale.

C Supplementary selection and model tables

Table 14: Training-only feature screening and OLS coefficients on the standardized scale.

Feature	Correlation_with_quality	VIF	OLS_standardized_coefficient	Stepwise_AIC_retained
alcohol	0.475	3.511	0.312	Yes
volatile_acidity	-0.369	1.829	-0.189	Yes
sulphates	0.280	1.508	0.179	Yes
citric_acid	0.220	3.225	-0.061	No
density	-0.177	6.993	-0.017	No
total_sulfur_dioxide	-0.157	3.335	-0.089	Yes
chlorides	-0.133	1.525	-0.088	Yes
fixed_acidity	0.116	7.903	0.042	No
ph	-0.050	3.409	-0.076	Yes
free_sulfur_dioxide	-0.047	3.011	0.058	Yes
residual_sugar	0.035	1.834	0.013	No

Table 15: Coefficients at minimum-CV penalties on the standardized transformed scale.

Feature	OLS	Ridge	Lasso	Elastic_Net
fixed_acidity	0.0424	0.0573	0.0159	0.0165
volatile_acidity	-0.1894	-0.1740	-0.1799	-0.1799
citric_acid	-0.0607	-0.0339	-0.0330	-0.0336
residual_sugar	0.0126	0.0209	0.0000	0.0000
chlorides	-0.0876	-0.0851	-0.0842	-0.0844
free_sulfur_dioxide	0.0577	0.0451	0.0357	0.0367
total_sulfur_dioxide	-0.0895	-0.0793	-0.0689	-0.0699
density	-0.0167	-0.0472	0.0000	0.0000
ph	-0.0756	-0.0525	-0.0707	-0.0708
sulphates	0.1792	0.1736	0.1691	0.1692
alcohol	0.3125	0.2789	0.3179	0.3175

Table 16: Elastic Net mixing-parameter search on shared folds.

Alpha	CV_RMSE_min	Lambda_min
0.25	0.6713	0.0131
0.50	0.6712	0.0093
0.75	0.6712	0.0065

Table 17: Holdout error by observed score for the pre-locked Ridge model.

Quality	N	Mean_Error	MAE	RMSE
3	2	-1.8537	1.8537	1.8576
4	11	-1.2038	1.2038	1.2421
5	115	-0.3276	0.3482	0.4496
6	107	0.1816	0.4030	0.4812
7	33	0.7812	0.7812	0.8252
8	3	1.5849	1.5849	1.5908

D Reproducibility and collaboration

The canonical analysis is rebuilt from RStudio with `source("analysis/00_run_all.R")`, or from a terminal with `Rscript --vanilla analysis/00_run_all.R`. The five workstreams execute in dependency order, every random operation uses a recorded seed, and `output/artifact_manifest.csv` stores file sizes and MD5 checksums. Row-level holdout outcomes are not serialized. Software versions for the verified run are listed below; full `sessionInfo()` is in `output/logs/session_info.txt`. The analysis and dynamic report use R, `knitr`, and `rmarkdown` (R Core Team 2026; Xie 2025; Allaire et al. 2026).

The workstream allocation and review interfaces are documented in the repository's collaboration work-split file. The final integration review verified the technical handoffs recorded in the contribution statement.

Table 18: Software versions used for the verified run.

Package	Version
R	4.5.2
glmnet	5.0
knitr	1.51
rmarkdown	2.31

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